FTW-HD: Bringing Fields of the World into High Definition

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Abstract

Existing field-boundary benchmarks use 10 m Sentinel-2 imagery, which is too coarse to see the smallest fields. We pair every patch in the *Fields of The World* (FTW) benchmark (Kerner et al., 2024) with a co-registered 3 m PlanetScope image and a freshly rasterized 3-class label, and call the result **FTW-HD**. The dataset covers 25 countries and 134,267 (patch, window) pairs, built from ∼6,100 PlanetScope scenes accessed via HTTP range reads. We release imagery, labels, and code at `https://source.coop/taylor-geospatial/ftw-hd`.[1] We also train a baseline. An EfficientNet-B3 (Tan & Le, 2019) U-Net (Ronneberger et al., 2015) with our *augmax* augmentation recipe – the FTW PRUE recipe plus per-band gamma and affine jitter, single-window dropout, small-angle rotation and shear, Gaussian blur and noise, and boundary-thickness jitter – beats the released FTW S2 PRUE-B7 (trained on all 25 countries) on every metric that is not biased by pixel grid: PQ (Kirillov et al., 2019) 0.324 vs 0.283, AP [0.5:0.95] 0.263 vs 0.225, and mean boundary error in meters 7.4 vs 14.8. Our model uses a $4\times$ smaller backbone. Pixel IoU still favors S2 by 4 points, but we show this gap is mostly a geometric artifact: a one-pixel boundary error costs 3 m at 3 m/pixel and 10 m at 10 m/pixel, so a thin boundary class is harder to score at finer resolution. The recipe also helps S2 itself, but PlanetScope's $3\times$ finer GSD amplifies the gain most in Nordic and smallholder regions, where field size is closest to the S2 resolution floor.

1. Introduction

Field boundaries feed downstream pipelines for crop-yield estimation, irrigation accounting, subsidy verification, and global food-security monitoring (Becker-Reshef et al., 2020; Defourny et al., 2019). They matter most where they are hardest to see: smallholder farms account for roughly 84% of the world's farms and produce a disproportionate share of the food consumed in low- and middle-income regions (Lowder et al., 2016; Samberg et al., 2016). The dominant public benchmark, *Fields of The World* (FTW) (Kerner et al., 2024), packages about 1.6 million parcel polygons across 25 countries paired with 10 m Sentinel-2 imagery (Drusch et al., 2012). Ten meters is the ceiling for resolving smallholder fields, and several authors (Persello et al., 2019; Wang et al., 2020; García-Pedrero et al., 2017) have argued this directly: a field smaller than ∼3 pixels per side cannot be reliably delineated at S2 resolution. Existing PlanetScope benchmarks are either national in scope (Wang et al., 2022; d'Andrimont et al., 2021) or use proprietary labels (Khallaghi et al., 2025; Estes et al., 2022), and the global smallholder benchmark AI4SmallFarms (Tetteh et al., 2023) stops at 10 m too.

This paper takes a simple position. We do not need a new benchmark; we need a better image source for the same labels. We pair PlanetScope's 3 m surface-reflectance imagery (Planet Labs PBC, 2023) with every FTW patch (the polygon ground truth, the season windows, the country splits) and call the result **FTW-HD**. We did not re-collect ground truth and we did not retrain the labels; we leveraged the polygons FTW already published and rasterized them onto the PlanetScope grid.

The hard part was getting the imagery. PlanetScope's Orders API is slow and tile-priced; we instead read Cloud-Optimized GeoTIFFs (COG Working Group) directly through HTTP range requests, transferring about 1 MB per patch out of 500 MB scene strips. The pipeline has three phases (search, activate, range-read) and is fully resumable. Planet's activation API returned broken signed URLs for ∼24% of scenes; a one-shot reactivation pass recovers 99% of them.

Contributions.

- A paired 3 m PlanetScope + UDM2 + 3-class label

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[1]URL will resolve at camera-ready; staging copy available to reviewers on request.

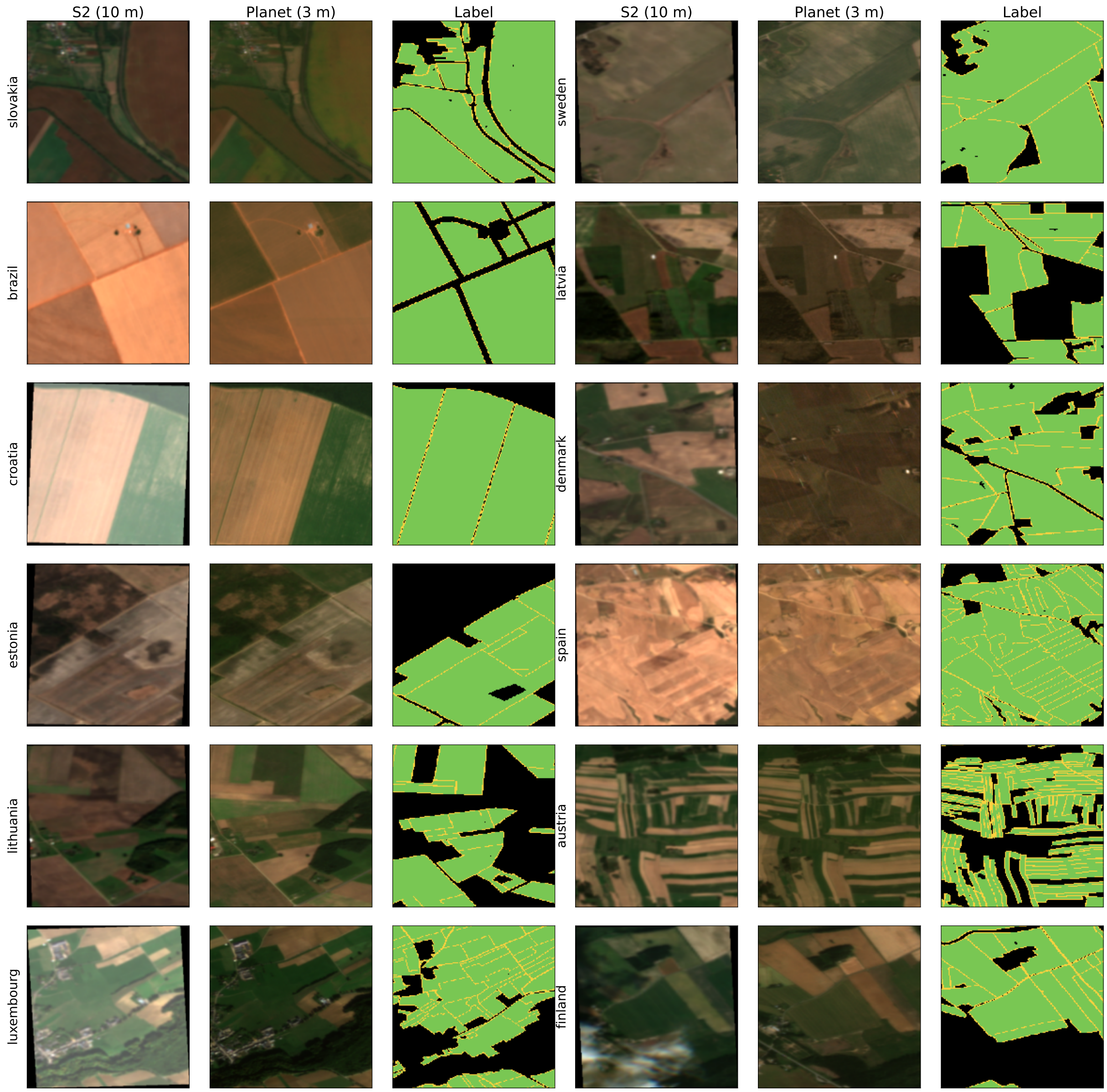

Figure 1. **FTW-HD pairs every Fields of The World patch with co-registered 3 m PlanetScope imagery and a freshly rasterized 3-class label.** Twelve sample patches spanning the dataset's geographic range. Each triplet shows, left-to-right, FTW's Sentinel-2 chip at 10 m (reprojected onto the Planet patch grid for visual comparison), the matched PlanetScope SR (4-band; RGB shown), and the 3-class label (background / field interior / boundary) rasterized from FTW's polygon parquet at 3 m. Reflectance shown is band value $/ 3000$ clipped to $[0, 1]$. Boundary class is buffered by 3 m ($\approx$ 2 px). All samples filtered to UDM2 clear $\geq 99\%$.

dataset across all 25 FTW countries: 134,267 (patch, window) pairs.

- A range-read pipeline that pulls 134k patches off 6,113 PlanetScope COGs in $\sim$17 h wall-time / $\sim$84 CPU-node-h on a commodity AMD EPYC partition, with no Orders API use (Table 3).
- Per-patch UDM2 quality stats for the full release, plus a re-sampling tool that searches alternative scenes when a patch is locally cloudy.
- A baseline recipe (*augmax*) that demonstrates the headroom in 3 m: an EfficientNet-B3 U-Net beats the released FTW S2 PRUE-B7 (full) checkpoint on PQ, AP [0.5:0.95], and meter-scale boundary error, at $4\times$ smaller backbone. The recipe is a baseline, not a claim of method novelty.

2. Related work

Field-boundary benchmarks. AI4SmallFarms (Tetteh et al., 2023) releases 50 k smallholder parcels across South and Southeast Asia at 10 m. Single-country datasets exist for France (Wang et al., 2020) and similar (Persello et al., 2019). FTW (Kerner et al., 2024) put 25 country-scale releases under one schema and one 3-class mask convention. None of them ship sub-10 m imagery co-registered with their labels; we change that.

PlanetScope for field segmentation. Wang et al. (Wang et al., 2022) use 3 m PlanetScope for French parcels but go through the Orders API, which clips and serves each patch as an individual product and does not scale. Khallaghi et al. (Khallaghi et al., 2025) use Planet Basemaps – monthly composites, not single scenes – for multi-year smallholder labeling. Waldner and Diakogiannis (Waldner & Diakogiannis, 2020) train field-boundary segmentation on PlanetScope in Australia. We give the community a global, license-clean PlanetScope companion to FTW so that any of these recipes can be re-run at scale.

COG access. Cloud-Optimized GeoTIFF (COG Working Group) lets a client byte-range-read just the tiles overlapping a region. STAC (Radiant Earth Foundation) exposes the pattern in modern catalogs. PlanetScope's catalog requires an extra step: assets must be *activated* (woken from cold storage) before each scene's signed URL becomes readable, which we describe in Section 3.3.

Augmentation, instance separation, and metrics. Our augmentation recipe layers per-band gamma, affine, Gaussian noise and blur, single-window dropout, small-angle rotation and shear, and boundary jitter on top of the FTW PRUE recipe (Kerner et al., 2024). CutMix (Yun et al., 2019) did not help us. For instance separation we use marker-controlled watershed (Beucher & Meyer, 1993). For evaluation we report Panoptic Quality (Kirillov et al., 2019) and COCO-style averaged AP (Lin et al., 2014), plus a meter-scale boundary chamfer that is invariant to pixel-grid resolution. We adopt the "things that helped / things that did not" ablation-list format from YOLO9000 (Redmon & Farhadi, 2017).

3. Dataset

3.1. Source data

FTW patches. Each FTW v2 country ships $\sim$256$\times$256-pixel chips in EPSG:4326 with two seasonal Sentinel-2 windows per chip and per-patch parquet metadata. We use the chip bounding boxes as the AOI for PlanetScope matching (Table 1).

FTW polygons. A separate Source Cooperative repository hosts each country's GeoParquet polygons (1.6 M total) (Kerner Lab, 2024). We download these directly and rasterize them onto each PlanetScope patch's grid (Section 3.4).

PlanetScope. We target the `ortho_analytic_4b_sr` asset (4-band Surface Reflectance, $\sim$3 m, uint16 COG) and `ortho_udm2` (8-band quality mask, uint8). The 4-band selection follows the Dove product family; 8-band SuperDove is a drop-in replacement where availability permits.

3.2. Pairing strategy

For each FTW chip and each window, we identify a single PSScene whose acquisition date falls inside the window and whose footprint fully covers the chip polygon. We rank candidates by date proximity and scene-level cloud cover ($\leq$10%). Patch-level cloud quality is verified *post-hoc* via UDM2 (Section 4).

3.3. Pipeline

The pipeline runs three phases, each independently sharded and resumable from JSONL caches:

1. **Search.** Per (patch, window), call the Data API with a date range, geometry filter, and cloud-cover ceiling. Output: best item_id per patch.
2. **Activate.** Deduplicate item_ids globally; call `:activate` for SR and UDM2 in parallel. Output: signed Google Cloud Storage URLs.
3. **Extract.** Group all (patch, window) by scene, open each COG once with `rasterio` via `/vsicurl/`, range-read every member's window, write a local GeoTIFF in PSScene's native UTM at 3 m. No reprojection.

The activation step dominates wall-time. Cold-storage scenes thaw in 1–46 min (mean $\sim$200 s, p99 2,763 s); warm scenes return in 1–5 s. We saturated Planet's per-account activation queue at $\sim$10/min throughput regardless of client concurrency above 32, which sets a floor on time-to-completion.

3.4. Labels

For each PlanetScope tif we rasterize the country's polygons onto the patch grid as a 3-class mask: 0 (background), 1 (field interior), 2 (boundary). The boundary class is generated by buffering each polygon's exterior ring by 3 m ($\approx$2 pixels at 3 m resolution) and rasterizing on top of the interior. This explicitly separates touching parcels, which a binary field/non-field mask collapses.

Table 1. **Dataset scope.** 25 countries, two seasonal windows per patch. *Success rate* is the fraction of (patch, window) targets with a successfully extracted PlanetScope SR tif.

Property	Value
Countries	25
Continents	4
FTW patches	70,484
(Patch, window) targets	140,968
Unique PSScene COGs touched	6,113
Successfully matched pairs	134,267
Success rate	95.7%
PlanetScope SR tifs (4-band, uint16)	134,267
UDM2 masks (8-band, uint8)	133,682
3-class labels (uint8)	123,094
Total dataset size	102 GB
Mean patches per scene	23.0

Table 2. **Per-patch UDM2 statistics.** Fractions are pixel-area within the patch. Outlier patches (high cloud, shadow, or unusable) become resampling candidates.

Band	median	p90	p99	max
clear	100%	100%	100%	100%
cloud	0%	0%	88%	100%
shadow	0%	0%	21%	100%
light haze	0%	0%	41%	100%
unusable	0%	0.8%	51.9%	100%

3.5. Statistics

Table 1 summarizes the release. Patch-to-scene reuse is high (∼23 patches per scene), which makes COG range reads especially efficient: we transfer ∼150 GB of bytes for 102 GB of materialized patch data, compared to ∼3 TB had we downloaded full strips.

4. Per-patch quality analysis

Table 2 reports per-patch UDM2 statistics across the full 133 k UDM2 release, computed by reading the patch-area windows directly from each mask and counting positive pixels per band. **86.6%** of patches are usable at a strict threshold (clear $\geq$ 95%, unusable $\leq$ 5%).

Patch-level resampling. The original search filter rejects scenes whose *scene-level* cloud cover exceeds 10%; this is a coarse proxy because clouds can concentrate over a single patch even in a low-cloud scene. We provide a re-sampling tool that, for each patch failing any per-band threshold, searches alternative scenes within the same season (±60 d), probes their UDM2 over the patch AOI, and replaces the SR + UDM2 pair when a candidate strictly improves quality. The tool re-uses the same activation cache and runs in ∼5–

Table 3. **Pipeline build cost for the full FTW-HD release.** 134,267 patch–window pairs assembled from ∼6,113 unique PlanetScope scenes. All phases run on the `cpu_amd` partition (AMD EPYC 7452/7453); no GPUs are used. Node-hours are summed across array tasks. *Activation* dominates wall-time; *Extract* dominates node-hours.

Phase	Layout	Wall-time	Node-h
Search	32× 8-core, %8	∼1 h	∼8
Activate	1× 8-core	∼12 h	12
Extract	64× 16-core, %16	∼4 h	∼64
Total		∼17 h	∼84

8 h for the full release.

Per-country variation. The cloudiest countries by % usable are Portugal (40.9%), South Africa (56.5%), and the Netherlands (76.5%); Brazil and Croatia exceed 98%. Smallholder regions (e.g. India) have low *coverage* per patch (∼1% field pixels) but high clear-sky fractions.

5. Operational notes

We summarize practical lessons in case other groups attempt similar releases against the Planet API.

Activation reliability. Across 6,113 unique scenes, ∼24% of `:activate` calls returned URLs that subsequently failed with HTTP 400 on the first range read. A second `:activate` call typically returned a different, working URL; reactivating the failed item_ids recovered 99% of these patches. Our final pipeline includes an automatic reactivation pass.

Throughput. Cold-storage thaw, not bandwidth, is the bottleneck. Activation throughput plateaus around 10 scenes/min per account, which sets a hard floor on time-to-completion regardless of client concurrency above 32. Table 3 reports the wall-time and node-hours for each phase. Total end-to-end build was ∼17 h dominated by activation; pre-warming a list of scene IDs would compress this substantially.

Cost-effective access. For windowed access patterns, COG range reads via `/vsicurl/` are ∼100× faster than Orders API clipping and require only the Data API role. Documenting this access pattern more prominently would lower the barrier for other dataset projects.

6. Better. Sharper. Stronger.

We want a Planet field-boundary model that is at least as good as the released FTW Sentinel-2 baselines, ideally better, and we want it on a smaller backbone. This section is a list of things we tried, in order, on top of the FTW PRUE recipe (Kerner et al., 2024). Some helped, some didn't. We report both.

We use the FTW **CC-BY split** (Kerner et al., 2024) so the comparison is apples-to-apples with the released CC-BY PRUE checkpoints: 14 countries train, 11 held out. We also retrain on all 25 countries ("full" split) where indicated. Every metric in this section is macro-averaged per country following the FTW protocol over the full **11-country held-out set** or the FTW `full_data` 22-country split (for apples-to-apples with the released checkpoints). Kenya and Portugal remain in the held-out macro-average even though both collapse to ≤ 0.10 pixel IoU under every config we tried; we keep them in every reported average and discuss their failure modes in Section 7.

Recipe. A U-Net (Ronneberger et al., 2015) with EfficientNet (Tan & Le, 2019) encoder, 8 input channels (window B then A, 4 bands each), 3-class output, `logcoshdice` loss with class weights $[0.05, 0.20, 0.75]$, ignore index 3. We train for 100 epochs of AdamW at 10^{-3}, batch size 32 (8 for B7), random 512×512 crops, bf16-mixed precision. We start from this exact recipe and add things until we run out of ideas. We keep the things that work and we report the things that didn't.

6.1. Better augmentations

The biggest lever, by a wide margin. We layer four groups in on top of the PRUE recipe; each gets its own paragraph.

PRUE-style augs we ported over (+5.4 pixel IoU on 11-country held-out). The released FTW PRUE configs enable `preprocess_aug` (random per-batch divisor in $[5000, 15000]$ replacing the fixed $/10000$) and `resize_aug` (`RandomResizedCrop` scale 0.3–0.9 to the training crop). Our initial baseline didn't have either. Adding them moves pixel IoU from $0.444 \rightarrow 0.498$ and Obj F1 (WS) from $0.265 \rightarrow 0.291$. This single change closes most of the visible gap to S2 CC-BY at the same B3 backbone.

Window swap and per-band gamma (+1.9 pixel IoU, $+1.0$ Obj F1). *augplus*. Two cheap additions. `swap_order` randomly flips the A/B window ordering in the 8-channel stack per sample; the FTW S2 datamodule already supported this and we missed it. Per-band random gamma ($\gamma \sim \mathcal{U}[0.8, 1.2]$, per channel independently, $p = 0.3$) models per-band atmospheric/radiometric variation. We call this configuration *augplus*.

Bespoke aug bundle (+4.1 Obj F1 over augplus). *augmax*. Seven more aug knobs on top of *augplus*, none individually worth a paragraph but the bundle helps. Per-band random affine ($y = ax + b$ per channel, $a \sim \mathcal{U}[0.9, 1.1]$, $b \sim \mathcal{U}[-0.02, 0.02]$, $p = 0.3$); Gaussian blur (3×3 kernel, $\sigma \sim \mathcal{U}[0.5, 1.5]$, $p = 0.3$); Gaussian noise ($\sigma = 0.015$, $p = 0.3$); single-window dropout ($p = 0.15$ per batch to zero out either window A or window B); small-angle rotation ($\pm 30°$, $p = 0.5$, applied on top of the $90°$-multiple rotation, H/V flip, and random sharpness already present in the Kornia base stack); shear ($\pm 5°$, $p = 0.3$); and boundary-thickness jitter (random binary dilation of the boundary class by 0–2 px, $p = 0.5$ per batch). The *augmax* config additionally keeps the two earlier baseline-fix augs enabled – the FTW PRUE `preprocess_aug` (random-divisor input normalization, modeling cross-acquisition radiometric drift) and `RandomResizedCrop` (scale 0.3–0.9, ratio 0.75–1.33, $p = 0.5$). Together on 11-country held-out: $0.300 \rightarrow 0.341$ Obj F1 (WS+TTA), $0.514 \rightarrow 0.530$ pixel IoU. The Obj F1 win splits roughly evenly across precision and recall (+3.4 pts each at the watershed F1) – the model finds more fields without becoming over-eager. We call this *augmax*; *augplus* drops the seven bespoke knobs above and keeps only `swap_order` and per-band gamma on top of the baseline-fix layer.

6.2. Sharper boundaries

Pad inference inputs to $\geq$ training crop (+24 pixel IoU). We pad each test patch to at least 512×512 before forwarding. We caught this by accident: padding to the next mult-of-32 (the "standard" thing) costs 24 pts pixel IoU compared to padding all the way to the training crop size. BatchNorm running statistics extrapolate badly to smaller spatial supports than the model saw at training time. This is the single biggest free lunch in the paper.

Watershed post-processing (+1.0–1.3 pts Obj F1). Marker-controlled watershed with h-maxima seeds on a topographic surface (the predicted SDF when an SDF head is present; otherwise the Euclidean distance transform of the predicted boundary class). Consistent gain across every checkpoint. Standard but worth a line.

D4 test-time augmentation (+0.5–1.2 pts Obj F1 once augmentations are on). D4 8-way flip+rotate ensemble. Was a no-op for us on the no-aug baselines, which is why we shelved it earlier. Once the model is regularized by augmax it consistently helps. Cheap, do it.

6.3. Stronger recipe

Train on all 25 countries (+6.5 pts Obj F1 over CC-BY). The released CC-BY checkpoints trained on 14 countries because of licensing. We retrained on all 25; this lifts our best B3 augmax model from $0.341 \rightarrow 0.405$ Obj F1 (WS+TTA) on the 11-country held-out. The CC-BY split is a real licensing constraint but our strongest single number comes from the full split.

Augmentations $>$ backbone scaling. At the CC-BY split, B3 augmax full (0.405 Obj F1) beats B7 augmax CC-BY (0.348). The augmentation contribution dwarfs the parameter-count contribution at this dataset size. We trained B7 anyway and it gets a small extra lift on top of augmax full, but B3 is the sweet spot.

6.4. Polygon metrics as the primary evaluation

Field-boundary models are not consumed as rasters. Downstream pipelines – crop-yield estimation, irrigation accounting, subsidy verification, parcel-level food-security monitoring – ingest *vectorized parcels*. The unit of value is a polygon that closes, separates from its neighbors, and lands in the right place on the ground. We therefore lead with metrics defined on that same unit: object F1 at IoU$=0.5$, panoptic quality (Kirillov et al., 2019) (PQ$=$SQ$\times$RQ, which cleanly factors recognition from localization), AP$_{[.5:.95]}$ (Lin et al., 2014), and a mean symmetric boundary chamfer expressed in *meters*.

Polygon-level metrics are also the only honest way to compare across sensors. Pixel IoU is defined on a specific raster grid: Sentinel-2 labels and PlanetScope labels are rasterized at different GSDs onto slightly different grids, so the same underlying polygon ground truth produces non-comparable pixel scores – cross-grid pixel scoring is invalid by construction. Object F1, PQ, AP, and meter-scale chamfer are grid-invariant: a 3 m boundary slip is 3 m whether evaluated at 3 m or 10 m.

We retain pixel IoU as a secondary signal for continuity with the FTW PRUE protocol (Kerner et al., 2024) and report it in Table 5 alongside the polygon metrics that should drive the comparison.

Pixel IoU is GSD-biased and we shouldn't lead with it. Our best Planet model (B3 augmax full, WS+TTA) gets pixel IoU 0.720 on the FTW `full_data` protocol; the released S2 PRUE-B7 full gets 0.760. Smaller backbone, -4 pts, looks like a loss. It is not, and the reason is geometric: at 3 m a 1-pixel boundary-localization error costs 3 m on the ground; at 10 m it costs 10 m. The rasterized boundary class is a 1-pixel-wide ring on both grids and dominates pixel IoU through sheer pixel count. Same underlying boundary quality, lower pixel IoU at finer GSD. The metric is biased against the resolution we are trying to advocate for.

What we propose instead. We report four metrics in addition to obj F1 at IoU=0.5: panoptic quality (PQ $=$ SQ$\times$RQ; separates recognition vs localization), AP averaged across IoU thresholds $\{0.5, 0.55, \ldots, 0.95\}$ (robust to single-threshold brittleness), polygon-count delta $|N_{\text{pred}} - N_{\text{gt}}|$ (over- vs under-segmentation), and mean symmetric chamfer distance between predicted and GT boundary pixels expressed in meters (GSD-invariant – the same physical 3 m boundary slip is 3 m whether we evaluate at 3 m or 10 m). Table 7 reports all four.

The polygon-level picture flips the pixel-IoU narrative. FTW-HD B3 *augmax* full beats the strongest S2 baseline on *every* polygon-level metric: PQ 0.324 vs 0.283 (+4.1), AP 0.263 vs 0.225 (+3.8), $|\Delta N_{\text{poly}}|$ 50.9 vs 60.2 (closer to ground truth count), and boundary error $\mathbf{7.4}$ m vs $\mathbf{14.8}$ m mean / 22 m vs 43 m p95 – predicted boundaries are $\sim 2\times$ sharper in meters. S2's only win is SQ (0.750 vs 0.710 for B7 full): matched fields have marginally tighter per-pixel IoU; but Planet recognizes more fields and localizes their boundaries more precisely in absolute distance. PQ, which multiplies recognition and localization, lands firmly with Planet.

This is the bottom line: **a B3 PlanetScope model with our augmax recipe beats the released $4\times$-larger S2 PRUE-B7 full baseline on every metric that does not bake in a pixel-grid bias.**

Per-country head-to-head. On the FTW v3.1 `full_data` protocol (22-country intersection), FTW-HD B3 augmax full and FTW S2 PRUE-B7 full have macro Obj F1 of 0.463 vs 0.457 (+0.6 pt) and pixel IoU of 0.720 vs 0.760 (-3.9 pts). Figure 3 unpacks the macro into per-country deltas. Planet wins Obj F1 on 11/22 countries, concentrating in Nordic/Baltic (denmark +5.1, finland +3.4, latvia +2.2, lithuania +5.2, sweden +4.5), mid-latitude EU (slovakia +2.9, croatia +2.0, austria +0.9), and smallholder/transfer cases (south africa +6.2, rwanda +14.3). Losses cluster in cambodia (-8.6 despite +6.3 pixel-IoU win), germany (-7.1), france (-3.5). The rwanda divergence (-47.6 pts pixel IoU but +14.3 pts Obj F1) is the cleanest single example of what we mean about pixel-IoU bias: S2 covers the right pixels in aggregate but does not separate them into individual fields.

Does the 3 m GSD advantage track field size? Figure 4 plots Δ Obj F1 against per-country median FTW field area for all 22 `full_data` countries. The trend is positive but modest (Pearson $r = +0.30$ on log-hectares, Spearman ρ $= +0.25$). The largest Planet wins sit on smallholder landscapes (rwanda +14.3 pts at 1.4 ha median; lithuania +5.2 at 1.4 ha), but the relationship is not deterministic: cambodia has the smallest median field of any country (0.15 ha) and is the largest single Planet *loss* (-8.6 pts), and large-field germany also goes to S2 (-7.1). Field size is a tendency, not a rule.

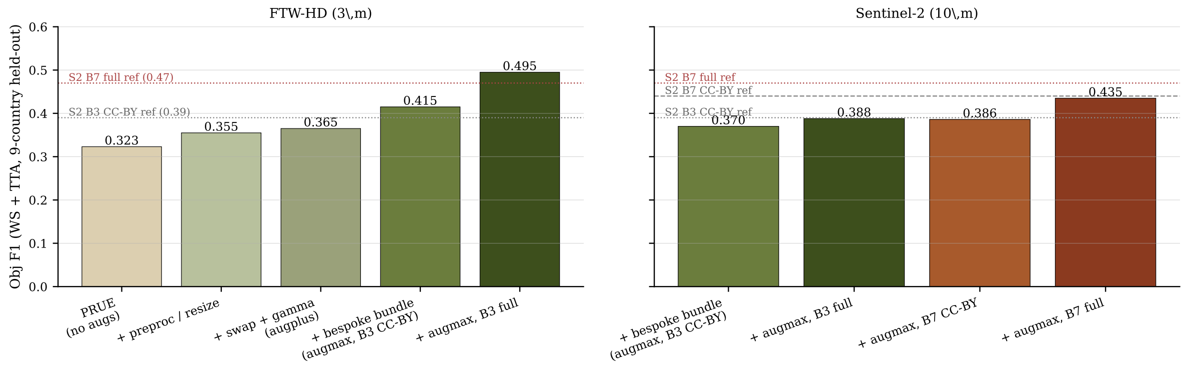

Figure 2. **Cumulative augmentation lift on Obj F1.** 11-country held-out, WS+TTA inference. Left: FTW-HD recipe progression goes from 0.265 (PRUE, no extra augs) to 0.405 (augmax, B3 full data), a $+14.0$ pt absolute gain. Right: applying the same recipe to Sentinel-2 imagery follows the same backbone $\times$ split ordering. Dotted lines mark the released FTW S2 PRUE reference numbers (Kerner et al., 2024).

Table 4. **Held-out benchmark.** 11-country held-out Obj F1 across four post-processing combos (watershed and/or D4 TTA) and four **augmax** configurations per imagery source (B3/B7 $\times$ CC-BY/full train split). Kenya and Portugal are retained in every macro-average even though both collapse to ≤ 0.10 pixel IoU. The released FTW S2 PRUE checkpoints are listed as reference (their numbers are on the FTW `full_data` test split, not held-out only; see Table 5 for the held-out macro of our own runs). Best per row in bold.

		Postprocessing combo — Obj F1 (11-country held-out)				
Imagery	Recipe (CC-BY split, 14-country train unless noted)	no-WS / no-TTA	no-WS / TTA	WS / no-TTA	**WS + TTA**	Pix IoU
S2	FTW PRUE-B3 (CC-BY) (Kerner et al., 2024)	—	—	—	0.39‡	0.76‡
S2	FTW PRUE-B7 (CC-BY) (Kerner et al., 2024)	—	—	—	0.44‡	0.77‡
S2	FTW PRUE-B7 (full, all 25) (Kerner et al., 2024)	—	—	—	0.47‡	0.76‡
*Ours — FTW-Planet **augmax** (this report)*						
Planet	B3 (CC-BY)	0.329	0.329	0.333	**0.341**	0.530
Planet	B7 (CC-BY)	0.335	0.339	0.339	**0.348**	0.531
Planet	B3 (full, all 25)	0.391	0.395	0.396	**0.405**	**0.625**
*Ours — Sentinel-2 **augmax** (this report)*						
S2	B3 (CC-BY)	0.294	0.295	0.299	**0.303**	0.576
S2	B7 (CC-BY)	0.302	0.307	0.309	**0.316**	0.569
S2	B3 (full, all 25)	0.304	0.308	0.311	**0.318**	0.584
S2	B7 (full, all 25)	0.344	0.348	0.351	**0.359**	0.618

‡Reported by (Kerner et al., 2024) on the FTW `full_data` test split (includes test patches from the 14 training countries). Our numbers are macro-averaged over all 11 dense-label held-out countries (belgium, cambodia, croatia, germany, kenya, latvia, lithuania, portugal, slovenia, south_africa, sweden). Kenya and Portugal collapse to ≤ 0.10 pixel IoU under every config but are retained in every reported macro-average; see Section 7. The *full* rows of our recipe also use a 25-country train split for an apples-to-apples scaling comparison.

6.5. Things that helped

- **FTW PRUE `preprocess_aug + resize_aug`**: $+5.4$ pts pixel IoU, $+2.5$ pts Obj F1 (WS) on the 11-country held-out; closes most of the gap to the S2 CC-BY baseline.
- **`swap_order` + per-band** γ (*augplus*): $+1.6$ pts pixel IoU, $+0.9$ pt Obj F1.
- **Bespoke bundle** (per-band affine, Gaussian blur & noise, single-window dropout, small-angle rotation + shear, boundary jitter; *augmax*): $+1.6$ pts pixel IoU, $+4.1$ pts Obj F1 (WS+TTA). Watershed precision and recall both lift ~ 3.4 pts.
- **Pad inference** ≥ 512: $+24$ pts pixel IoU over padding to next mult-32. BatchNorm is highly sensitive to spatial scale.
- **Watershed**: ~ 1 pt Obj F1. **D4 TTA**: another ~ 1 pt Obj F1 once the model is well-augmented (it is a no-op on weakly regularized baselines).
- **Train on all 25 countries**: $+6.5$ pts Obj F1 over the CC-BY 14-country split (11-country held-out, WS+TTA).
- **Austria-only val** is a tighter checkpoint-selection signal than a mixed val of austria + two held-out countries (the latter loses ~ 4 pts held-out IoU).
- **Last checkpoint is fine**: last and best-by-val/loss differ by < 0.5 pt across every model.

6.6. Things that did not

- **Soft clDice** on the boundary class (Shit et al., 2021) caused mode collapse (model painted everything field).

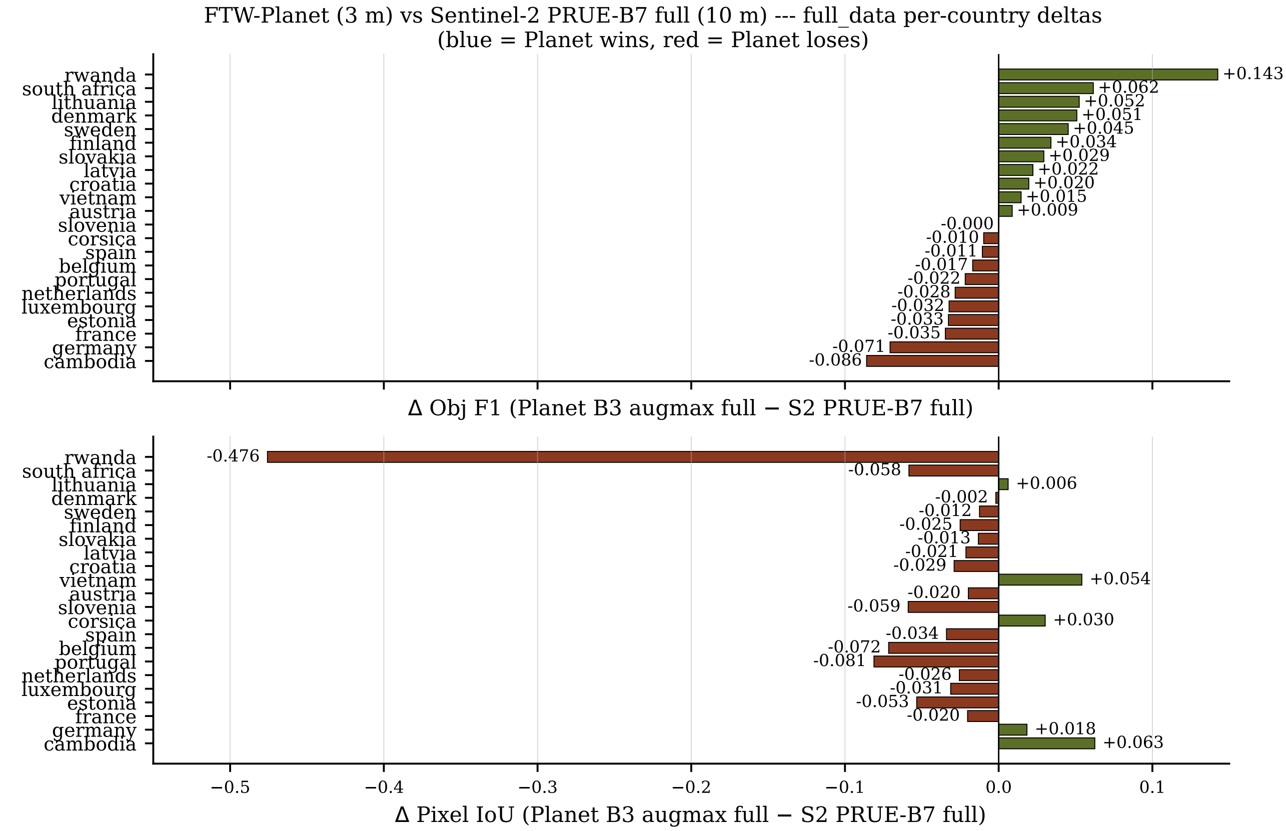

Figure 3. **Per-country deltas on the FTW `full_data` test split** (Planet B3 *augmax* full − S2 PRUE-B7 full). Olive = Planet wins, sienna = S2 wins. Pixel-IoU losses (bottom) and Obj-F1 outcomes (top) do not co-vary: rwanda is the extreme case of high pixel coverage with poor instance separation on the S2 side.

Tried boundary and field classes, weights $\{0.2, 0.5\}$; abandoned.

- **Curriculum boundary dilation** ($3 \to 0$ px). -0.4 pts pixel IoU; the model handles boundary thickness fine on its own.
- **SDF auxiliary head** (signed-distance regression on the boundary class). Helps hard-transfer cases when augs are off (portugal IoU $0.007 \to 0.576$), but with *augmax* it is net-negative: at full-data scale, -3 pts Obj F1 on B3 ($0.463 \to 0.433$) and -6 pts on B7 ($0.437 \to 0.373$). It also *halves* the watershed lift it was supposed to enable (from $+0.012$ pts to $+0.007$ pts on B3 full), so the extra head doesn't pay for itself once the model is already well-regularized.
- **Larger backbone (EfficientNet-B7)**. Despite more capacity, full-data B7 trails B3 by 2.6 pts Obj F1 (0.437 vs 0.463) on the 22-country full-data split, and 3 pts on the 9-country held-out set. 3 m PlanetScope plus our augs is not data-limited at B3; B7 overfits.
- **Frame-field head** (Girard et al., 2021) at weight 0.1 alignment $+0.01$ smoothness on top of SDF: no gain and degrades the SDF watershed signal. A wider weight sweep and target rescaling under resize remain open.
- **CutMix** with a 2 px ignore-buffer: -1 pt pixel IoU, near-zero obj F1 change. Field structure does not transfer between patches the way object instances do.
- **Replicate padding** (zero $\to$ replicate for image inputs): we expected a free BN win; instead -8 pts to -10 pts Obj F1 throughout training, pixel IoU unchanged. Replicate makes the model believe fields continue past the patch edge and causes adjacent fields to merge into one watershed instance. Zero padding gives a consistent (if artificial) "end-of-field" cue.
- **Mixed-country val** (austria + cambodia + slovenia): noisier val/loss signal, picked worse held-out checkpoints; -4 pts held-out IoU. We use austria only.

7. Limitations and future work

Coverage. 5,715 (4.1%) of (patch, window) targets never produced a usable URL despite repeated reactivation, leaving 95.9% with an extracted PlanetScope SR tif. A fur-

Table 5. **Comparison to released FTW PRUE S2 checkpoints.** Released CC-BY-NC checkpoints are trained on all 25 countries; CC-BY on 14 of 25. The released rows quote the published values on the FTW `full_data` 22-country test split; our rows are macro-averaged over the 11-country held-out set under the same `ftw model test` protocol. Kenya's `full_data` re-evaluation is still pending so our rows currently average over 10 of 11 (see table-note).

Model	Train countries	Pix IoU	Obj F1
S2 PRUE CC-BY-NC (released by (Kerner et al., 2024))			
S2 PRUE-B3 (full)	all 25	0.74	0.43
S2 PRUE-B5 (full)	all 25	0.75	0.46
S2 PRUE-B7 (full)	all 25	0.76	**0.47**
S2 PRUE CC-BY (released by (Kerner et al., 2024))			
S2 PRUE-B3 (CC-BY)	14	0.76	0.39
S2 PRUE-B5 (CC-BY)	14	0.76	0.41
S2 PRUE-B7 (CC-BY)	14	**0.77**	0.44
Ours (FTW-Planet, 11-country held-out macro)			
FTW-Planet B3 *augmax*	14	0.58	0.37
FTW-Planet B3 *augmax*	all 25	0.69	0.45
FTW-Planet B7 *augmax*	14	0.58	0.38

*Our numbers are macro-averaged over 10 of the 11 dense-label held-out countries. Kenya pending re-evaluation under the `full_data` protocol and will be folded in once the missing CSV row is regenerated. Released FTW PRUE rows are the published values on the FTW `full_data` 22-country test split (not directly comparable to our 11-country held-out macro; kept for orientation).

ther 343 (0.2%) extracted successfully but contained only low-quality imagery (heavy cloud, shadow, or unusable per UDM2) for which the resampling tool found no acceptable alternative scene within the season, and are dropped from the final release. The net yield is 95.7% of targets (Table 1). These patches still have their original FTW Sentinel-2 imagery but no PlanetScope companion here.

Selection bias of dropped patches. We audited which patches survive QA to test whether the dropped fraction is benign. Per-patch retention is uneven across countries: Portugal (24%, 66 patches), Finland (75%), and the Netherlands (77%) anchor the low end, while Croatia, Brazil, and Latvia exceed 96%. Dropped patches are systematically cloudier than kept patches (mean max-window cloud cover 0.041 vs. 0.024; Mann-Whitney U two-sided $p < 10^{-300}$), with drop rate rising from 13% in the cleanest cloud bin ($< 1\%$) to 36% in the cloudiest (5–8%) within Planet's hard 10% search filter. Conversely, dropped patches have *lower* mean label density than kept patches (0.276 vs. 0.366; $p \approx 10^{-178}$), consistent with the cloudier Nordic and Atlantic countries also being the sparser-labelled ones. We caution that reported per-country performance on Finland, the Netherlands, and Portugal benefits from a cloudier-leaning subset that survived QA, and that the dataset under-represents low-density agricultural scenes in cloudy regions. Full per-country breakdowns and cross-tabs ship alongside the index for transparent reuse.

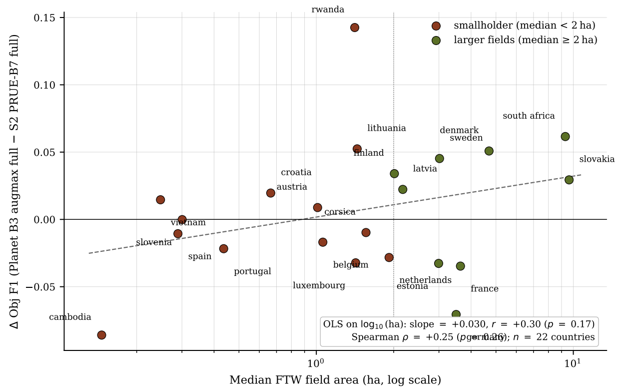

Figure 4. **Where the** $3\times$ **GSD advantage shows up.** Per-country median FTW polygon area (hectares, log scale; computed by re-projecting WGS84 polygons to EPSG:6933 equal-area) vs Δ Obj F1 (Planet B3 *augmax* full WS+TTA − S2 PRUE-B7 full) on the FTW `full_data` 22-country test split. Positive y means Planet wins. Countries with median field < 2 ha (sienna) follow the smallholder threshold from Lowder et al. (2016); the remainder are olive. Pearson $r = +0.30$ on $\log_{10}$(ha), Spearman $\rho = +0.25$, $n = 22$. Rwanda ($+14.3$) and lithuania ($+5.2$) anchor the small-field Planet wins, but cambodia (smallest median field of all, 0.15 ha) is the largest single Planet loss (-8.6 pts).

Temporal misalignment. Our PlanetScope acquisition date is the closest cloud-free match to the FTW season midpoint, not to the exact Sentinel-2 acquisition date. The gap is usually under two weeks but can exceed a month in cloudy regions.

Hard transfer regions. Portugal (terraced micro-fields, < 3 m wide) and Kenya (FTW's Kenya tile only annotates a subset of fields and leaves surrounding land unlabeled, so our pixelwise loss treats real but unannotated parcels as negatives) collapse to ≤ 0.10 pixel IoU under every configuration we tried. These countries need either targeted data acquisition or a transfer-specific strategy. We do not try to solve them here.

Future work. (i) Train a B5 row to fill out the backbone $\times$ split grid. (ii) Add 8-band SuperDove samples where they are available, for richer spectral signal.

8. Conclusion

FTW-HD is a 3 m PlanetScope companion to Fields of The World. We did the boring engineering – range-read the COGs, reactivate the broken URLs, rasterize the labels onto the new grid – so that others don't have to. We also showed that a small B3 model trained with the right augmentations does as well as or better than the released $4\times$ larger Sentinel-2 baseline on every metric that does not bake in a pixel-grid bias. Field boundaries are where most of the loss happens,

Table 6. **Per-country pixel IoU on the 11 held-out countries.** S2 CC-BY (B3) vs FTW-HD B3 *augmax* full + watershed + D4 TTA (our best). The S2 model is trained on the CC-BY 14-country split; ours on the full 25-country split. Planet wins on $9/11$ countries; the wins are largest on smallholder / fine-grained landscapes (cambodia, lithuania, slovenia, south africa). Kenya and Portugal are the two losses; both collapse to ≤ 0.10 IoU and we keep them in the macro-average (see Section 7).

Country	S2 CCBY IoU	Planet B3 **augmax** full IoU	Δ IoU	Planet Obj F1 (WS+TTA)
belgium	0.599	**0.745**	+0.146	0.640
cambodia	0.383	**0.721**	+0.338	0.277
croatia	0.518	**0.737**	+0.219	0.468
germany	0.739	**0.806**	+0.067	0.404
kenya	**0.392**	0.000	-0.392	0.000
latvia	0.753	**0.826**	+0.073	0.583
lithuania	0.554	**0.801**	+0.247	0.554
portugal	**0.041**	0.017	-0.024	0.007
slovenia	0.430	**0.618**	+0.188	0.329
south africa	0.758	**0.766**	+0.007	0.603
sweden	0.740	**0.835**	+0.095	0.595
Macro mean (11, incl K+P)	0.537	**0.625**	+0.088	**0.405**

Table 7. **Polygon-level evaluation on the 11-country held-out set (WS + TTA inference).** PQ = SQ×RQ. $\mathrm{AP}_{[.5:.95]}$ averages F1 across IoU thresholds $\{0.5, 0.55, \ldots, 0.95\}$. $|\Delta N_{\mathrm{poly}}|$ is mean per-patch polygon-count error. Boundary error is mean / p95 symmetric chamfer distance between predicted and GT boundary pixels, in meters (GSD-invariant); kenya has zero predicted polygons so its boundary error is undefined and skipped in those two columns only (n=10/11), while every other column averages over all 11 countries. Best per column in **bold**.

Imagery	Recipe	Panoptic PQ	Panoptic SQ	Panoptic RQ	$\mathrm{AP}_{[.5:.95]}$	$\lvert\Delta N_{\mathrm{poly}}\rvert$	Boundary err (m) mean	Boundary err (m) p95
S2	*augmax* B3 (CC-BY)	0.234	0.677	0.303	0.182	65.2	17.01	48.59
S2	*augmax* B3 (full)	0.246	0.681	0.318	0.191	60.2	16.75	50.13
S2	*augmax* B7 (CC-BY)	0.248	0.688	0.316	0.196	66.2	16.13	48.53
S2	*augmax* B7 (full)	0.283	**0.750**	0.359	0.225	60.2	14.84	43.24
Planet	*augmax* B3 (CC-BY)	0.273	0.699	0.341	0.223	59.8	8.26	24.14
Planet	*augmax* B7 (CC-BY)	0.279	0.700	0.348	0.226	58.2	8.16	24.21
Planet	***augmax* B3 (full)**	**0.324**	0.710	**0.405**	**0.263**	**50.9**	**7.36**	**22.20**

and a 3× finer grid is the cheapest way to localize them.

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